

Decentralized Computing and Blockchains 2025/26

Project 3 – DEX-NFT Pawning Dapp

Version 1.0

This project aims to implement a Dapp (web interface + smart contracts + server state) for managing ERC20 [1] (fungible) and ERC721 [2] (non-fungible) tokens. The Dapp must implement an NFT marketplace and allow users to manage DEX (a fungible token) and lend ether using DEX and NFTs as collaterals.

Project Description

This time, there's no detailed specification for the project. Below there's a list of required functionalities which the groups are free to implement them in the way they wish, as long as they can defend their choices in the end.

- 1) DEX marketplace: the Dapp must allow users to **buy** and **sell** DEX tokens from the Dapp smart contract using ether.
- 2) Ether lending using DEX as collateral: just like in project 2, a user can lend up to half of the value of a DEX collateral in ether. The user pays an interest value each payment cycle until returning the loan and the loan can be finalized early by paying a fixed tax.
- 3) NFT marketplace: the Dapp must allow users to **create** (mint) and **destroy** (burn) NFTs. Moreover, it allows users to **buy** and **sell** NFTs from other users (the Dapp doesn't own NFTs) using ether and DEX tokens (as required by the seller, but the Dapp can make the conversion if needed).
- 4) NFT auctions: users can mark their NFTs for selling for a fixed price or in an auction, with a minimum price and a maximum wait time.
- 5) Ether lending using NFT as a collateral: users can ask for a loan by providing an NFT (instead of DEX) as collateral. Since the Dapp doesn't own NFTs, this loan is only possible if another user provides the DEX to back the loan, getting the NFT from the user if it doesn't pay. In this type of loan, the user providing the DEX gets half of the interest or fixed tax paid by the lender.
- 6) Administrator console: there must be an administrator console for the smart contract owner define parameters of the Dapp.

Additional Remarks

Some additional remarks:

- All the security of the system must be anchored on the smart contracts;
- Although the project can be implemented in a single big smart contract, it is recommended to have at least two contracts (one for DEX, another for NFTs).

- Although it is possible to implement the work with only the web front end the smart contracts, it is advised to implement some server state to allow the management of additional information about users and NFTs.
- The project must be done based on techniques and software discussed in previous classes (smart contract programming libraries, hardhat, etc.).

Project Delivering

The delivery of the project should be a zip file containing:

- The identification of the group and students
- All the code for the project
- Instructions for running the code
- A report with up to 1000 words plus figures explaining the architecture and additional details about the functionalities of the Dapp.

Deadline

The project must be delivered on the course Moodle on May 27th (13:00hs).

References

- [1] <https://ethereum.org/en/developers/docs/standards/tokens/erc-20/>
[2] <https://ethereum.org/en/developers/docs/standards/tokens/erc-721/>